

where $\mathbf{1}$ is the indicator function (equal to 1 if true and 0 otherwise). Here, $\tilde{u}$, $\tilde{c}$, u , c , κ and d are strictly positive parameters, which we discuss below.

- (i) We normalize payoffs following ignore, $\mathcal{I}$, to $U_i = 0$.
- (ii) Payoffs from dislike, $\mathcal{D}$, depend on whether the article contains misinformation. We assume, in particular, that disliking has a cost of $\tilde{c} > 0$, regardless of whether the article is truthful (because of, say, the effort required to actively call out misinformation). In addition, disliking an article containing misinformation has a benefit of $\tilde{u} > \tilde{c}$, because individuals like calling out misleading articles. This formulation implies that disliking is never preferred to ignoring for an article that is truthful with probability 1, and is always preferred to ignoring for an article that contains misinformation with probability 1.
- (iii) Following a decision to share, $\mathcal{S}$, an agent receives utility from two sources. First, agents receive utility from sharing truthful content, but incur a cost from sharing misinformation. This explains the first component of utility following $\mathcal{S}$, $U_i^{(1)} = u\mathbf{1}_{\nu=\mathcal{T}} - c\mathbf{1}_{\nu=\mathcal{M}}$. Second, agents enjoy positive feedback from their peers (such as likes, or in our setting re-shares), but are negatively affected by dislikes. This is captured by the second component of utility $U_i^{(2)} = \kappa S_i - dD_i$. In this formulation, the parameter κ captures the importance of “popularity” for the agent’s sharing decision, while $-d$ represents the extent to which she cares about negative reactions.⁹

Information Structure and Solution Concept. Agents are not aware of when the article was first introduced onto social media, the prior sharing process, or the structure of the social network (though the link matrix $\mathbf{P}$ is common knowledge).¹⁰ Moreover, while any agent i knows the distribution $\{H_i\}_{i=1}^N$ of beliefs in the population, she does not know any agent j ’s belief (ideology) b_j . We focus on Bayesian-Nash equilibria, and refer to these as “equilibria” for short.

To eliminate trivial and unrealistic equilibria, we assume that the sensationalism of an article is upper bounded by $\bar{\kappa} = (c\tilde{c} - u(\tilde{u} - \tilde{c})) / (\tilde{u}N)$. This assumption guarantees that there is never an equilibrium where *every* agent *always* shares *all* articles. It also eliminates equilibria where agents may share and dislike, but never ignore.

Discussion — The basic assumptions introduced above are consistent with salient patterns of behavior and information structure in social media. First, as documented in studies such as Pennycook et al. (2021), users want to share content they believe to be truthful and not contain misinformation. Second, while users derive value from peer encouragement and re-shares on social media (Eckles et al. (2016)), they also suffer reputational costs when they get called out for sharing misinformation (see,

⁹Reputational loss associated with dislikes can be micro-founded straightforwardly by assuming that there are two types, normal and careless, and normal agents would like to distinguish themselves from careless ones who always share. We do not pursue this path, since imposing exogenous reputation costs simplifies the notation and analysis, with no major cost in terms of insights.

¹⁰More formally, we assume that all agents have uniform priors over these objects. In this sense, agents do not know the exact interactions and sharing patterns outside their neighborhood, which is consistent with the evidence in Breza et al. (2018). That being said, because the equilibrium sharing process is Markovian, this assumption can be relaxed by replacing $\mathbf{P}$ with the adjacency matrix.